Nikkila Prakash

Edinburgh, UK | nikkilaprakash.phy@gmail.com | LinkedIn | Portfolio | Willing to relocate

Education

University of Edinburgh — Edinburgh, UK
MSc Computer Science — *Sep. 2026 – Aug. 2027*

- Modules: Distributed Systems, Machine Learning, Deep Learning, AI & ML Systems Deployment
- Societies: CompSoc, QuantSoc, EdinburghAI, EU-Badminton

Experience

Software Engineer — Jul. 2023 – Jul. 2026
PayPal | Java, Spring Boot, Microservices, Kafka, BigQuery, Python, AMQ, RAG, MCP, AI

- Designed and developed a low-latency decisioning service that handles **10M requests/day**, enforcing compliance policies across the PayPal ecosystem.
- Developed an automated compliance validator using RAG for policy interpretation, reducing legal processing time by **60%**.
- **Technical owner** for **17 consent types**, leading 2 engineers; designed reusable frameworks that cut requirement onboarding **time by 40%**.
- Built an LLM-powered conversational assistant for privacy queries using OpenAI API and the Model Context Protocol (MCP), enabling self-service and cutting ticket resolution time by **15%**.
- Built core components of a consent management platform serving 400M+ users across PayPal, Venmo and Honey in the Americas, Europe and APAC, with real-time policy evaluation across jurisdictions.

Software Engineering Intern — Aug. 2022 – Jan. 2023
PayPal | Scala, Apache Spark

- Built a web crawler automation that classified cookies across **500K+ URLs** for European GDPR compliance, improving system efficiency by **30%**.

Projects

Low-Latency Exchange Simulator | *C++*

- Building a C++ exchange simulator with a limit order book and price-time priority matching engine supporting limit, market, IOC, cancel/replace orders.

High-Performance GPU Backtesting Engine | *CUDA, C++, Python, PyTorch*

- Developed a CUDA backtesting engine that maps each strategy to a GPU thread, parallelising path-dependent strategies across a **15,232-configuration** parameter grid over 10,000 minute bars in **34 ms**.
- Delivered a **12,000**× speedup over a naive Python baseline, **130**× over vectorised PyTorch and **14**× over single-threaded C++, with bit-identical P&L validated across all four implementations.

ML-Based Equity Trading Algorithm | *Python, scikit-learn, pandas, Alpaca API*

- Developed a systematic equity strategy on 20 years of daily data for 40 US equities (**190K observations**), pairing research-backed momentum and reversal factors with a random forest under embargoed validation.
- Showed that randomly shuffled train/test splits leak future data, overstating predictive power (daily IC) 10× versus time-ordered splits. Backtested the portfolio net of 5 bps costs and paper-traded it via Alpaca.

Patents & Publications

Autonomous Multi-Agent System for Production Errors — *Patent filed, 2025*

- LLM-powered multi-agent workflow that autonomously detects, classifies and resolves production errors.

Cardiovascular Disease Risk Assessment Using Machine Learning — *IEEE Xplore, ICICT 2023*

- First-author paper applying supervised learning to cardiovascular risk prediction. IEEE Xplore

Technical Skills

Languages: Java, Python, C++, Scala, SQL, JavaScript, HTML/CSS, XML, VBA
AI: LLMs, RAG, Multi-Agent Systems, Model Context Protocol (MCP), PyTorch, LangChain, HuggingFace
Backend & APIs: Spring Boot, JDBC, REST APIs, GraphQL, Microservices
Cloud & Infrastructure: AWS (EC2, S3, RDS, Lambda, Aurora), GCP, Docker
Databases: MySQL, MongoDB, Oracle DB, BigQuery, JunoDB
DevOps & Testing: Git, Jenkins, CI/CD, SonarQube, JUnit, Test-Driven Development (TDD), Agile

Achievements

- **Code.AI Hackathon, 1st place:** multi-agent privacy compliance system combining LLM policy interpretation with automated validation.
- **12 PayPal recognition awards** for innovation, technical excellence and collaboration.
- **Technical mentor** to 5 junior engineers (3 interns, 2 graduates) through onboarding, code reviews and career development.